

Austin Tan

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EDUCATION

New York University (NYU)

Bachelor of Science in Computer Science (GPA: 3.92/4.0)

New York City, NY

Expected May 2028

- **Relevant Coursework:** Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Object-Oriented Programming, Machine Learning, Artificial Intelligence, Computer Architecture, Databases, Probability & Statistics, Discrete Mathematics
- **Activities:** Tech@NYU Dev Team - Spring 2025

TECHNICAL SKILLS

Languages: C/C++, Python, Go, Java, TypeScript, JavaScript, SQL, Bash

Frameworks & Libraries: React, Next.js, Node.js, Pandas, NumPy, scikit-learn, Matplotlib, SQLAlchemy

Systems & Infrastructure: Linux, Docker, Kubernetes, PostgreSQL, Redis, CMake, GoogleTest, GCP, AWS

Developer Tools: Git, CI/CD, Jenkins, Claude Code, Codex, Cursor

WORK EXPERIENCE

Amazon Web Services (AWS)

Incoming Software Development Intern

Arlington, VA

Fall 2026

MathWorks

Software Engineering Intern - MATLAB Language QE Team

Natick, MA

May 2026 - Present

- Migrated 5 legacy MATLAB language test suites from a deprecated testharness framework to the xUnit-based MATLAB Unit Testing Framework, using automated conversion and manual refactoring to satisfy Code Analyzer standards and ship through CI.
- Owned test strategy for a dynamic built-in class update feature in MATLAB's Core Object System (MCOS); implemented C++ built-in classes/functions and built MATLAB test suites covering all downstream behavior

New York University

Undergraduate Research Assistant - Distributed Systems & Networking

New York City, NY

Apr 2026 - Present

- Developing a novel wide area network (WAN) consensus protocol in Go under Prof. Jinkun Geng that eliminates per-request inter-replica coordination overhead via predictive dispatch scheduling, enabling low-latency replication across geographically distributed machines
- Benchmarked the protocol on multi-region GCP VMs (US, Europe, Asia, South America) and bare metal CloudLab clusters (multiple US regions); built a custom TCP networking layer to measure WAN latency under background traffic, outperforming existing consensus protocols (Paxos, EPaxos) in throughput and latency

CrowdVolt (YC W24)

Software Engineering Intern

New York City, NY

Oct 2025 - Apr 2026

- Designed and implemented a full-stack price alert feature used 4000+ times across the marketplace; built PostgreSQL schema, REST APIs, and Next.js/React frontend, supporting end to end production deployment, monitoring, and reliability
- Built and deployed an AI data aggregation pipeline in Python using LLMs to normalize 1000+ event requests across artist, location, and date; automated team triage and surfaced results through an internal operations dashboard, significantly reducing manual review time
- Built a Redis-backed connection error triage service across a multi-stage bank verification pipeline, increasing seller bank linking conversion; included admin tooling to unblock stalled verifications
- Engineered an internal full-stack project management platform used daily by 15+ employees, with tooling for shift scheduling, analytics dashboards, transaction monitoring, and cart validation that reduced marketplace manipulation to near zero

SELECTED PROJECTS

Distributed Storage System | C++, liburing, Linux, Docker, CMake, GoogleTest

- Built a high-performance distributed append only log in C++20; designed a custom binary wire protocol with packed structs, network byte-order encoding, and sequence-numbered UDP ACKs with producer side timeout/retransmit; built a CLI client for producing and consuming messages across topics in a multi-node network
- Implemented the persistence layer using io_uring for high-throughput async I/O; stored records in a CRC32 validated binary format with crash recovery and partial write truncation; ensured multi-threaded safety across concurrent writers; verified correctness with 40+ unit and concurrency tests via Google Test
- Scaled the system horizontally by partitioning topics across 3 independent replica groups; implemented Network-Ordered Paxos (NOPaxos) with a sequencer to order log entries, enabling replication without replica coordination; achieved 0.53 ms median latency and 1,900 ops/sec sustained throughput; built a health monitor that tracks node liveness and triggers automatic failover for fault tolerance

Personalized Music Recommendation System | Python, NumPy, Pandas, scikit-learn, Matplotlib, Seaborn, Gradio

- Engineered a hybrid personalized recommendation system using collaborative filtering combining SVD (optimized with SGD for 20 epochs) and KNN, improving Precision@5 by 150% (0.036→0.091) and reducing training MSE by 60% (1.08→0.43)
- Processed 166K+ Last.fm interactions across 76K+ songs; matched 114K+ Spotify tracks for feature engineering and audio feature extraction with Z-score normalization and cosine similarity; evaluated ranking quality of SVD, KNN, and hybrid models using Precision/Recall/NDCG@k across an 84% sparse user-item matrix